Properties of DFT

Dr. Chayan Bhawal

EEE IIT Guwahati

EE 322M – Module-12b

Periodicity

Notation to denote N-point DFT pair $x[n]$ and $X[k]$

$$x[n] \underset{N}{\overset{DFT}{\Longleftrightarrow}} X[k]$$

$$x[n+N] = x[n] \text{ for all } n \qquad \text{and} \qquad X[k+N] = X[k] \text{ for all } k.$$

Proof: Define $W_N := e^{j\frac{2\pi}{N}}$.

$$x[n+N] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-k(n+N)} \quad \text{(From equation (15) Module 12a)}$$

$$= \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-kn}W_N^{-kN} = \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-kn} = x[n].$$

$$X[k+N] = \sum_{n=0}^{N-1} x[n]W_N^{k(n+N)} \quad \text{(From equation (14) Module 12a)}$$

$$= \sum_{n=0}^{N-1} x[n]W_N^{kn}W_N^{kN} = \sum_{n=0}^{N-1} x[n]W_N^{kn} = X[k].$$

Linearity

$$x_1[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_1[k] \qquad \text{and} \qquad x_2[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_2[k]$$

$$a_1 x_1[n] + a_2 x_2[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} a_1 X_1[k] + a_2 X_2[k]$$

Proof: Let the DFT of $a_1 x_[n] + a_2 x_2[n]$ be denoted by $X[k]$. Then

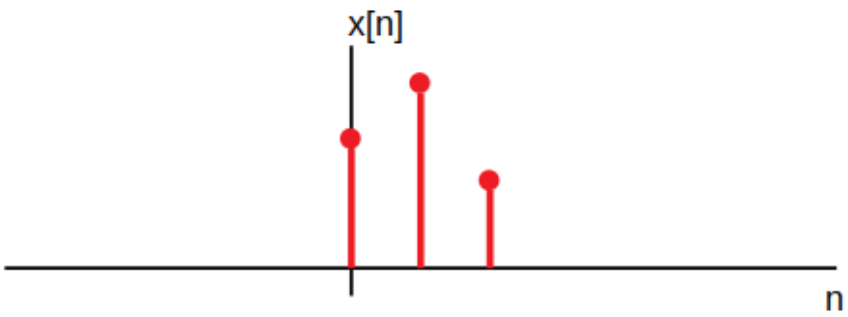
$$\begin{aligned}
X[k] &= \sum_{n=0}^{N-1} \left(a_1 x_1[n] + a_2 x_2[n]\right) W_N^{kn} \\
&= \sum_{n=0}^{N-1} a_1 x_1[n] W_N^{kn} + \sum_{n=0}^{N-1} a_2 x_2[n] W_N^{kn} \\
&= a_1 \sum_{n=0}^{N-1} x_1[n] W_N^{kn} + a_2 \sum_{n=0}^{N-1} x_2[n] W_N^{kn} \\
&= a_1 X_1[k] + a_2 X_2[k].
\end{aligned}$$

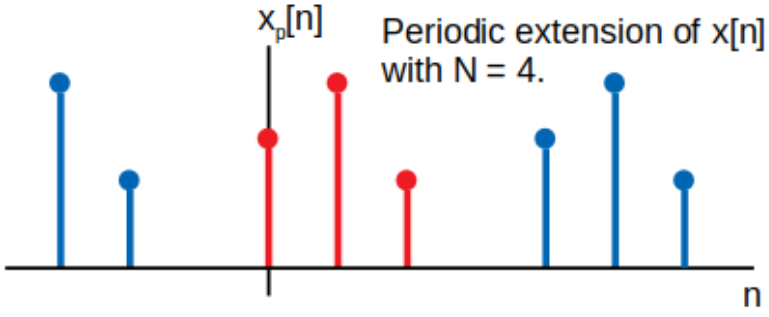
Circular symmetries

We have seen that N-point DFT of a finite duration sequence $x[n]$ of length $L \leqslant N$ is equivalent to the N-point DFT of a periodic sequence $x_p[n]$ of period N, which is obtained by periodically extending $x[n]$, i.e.,

$$x_p[n] = \sum_{\ell=-\infty}^{\infty} x[n - \ell N].$$

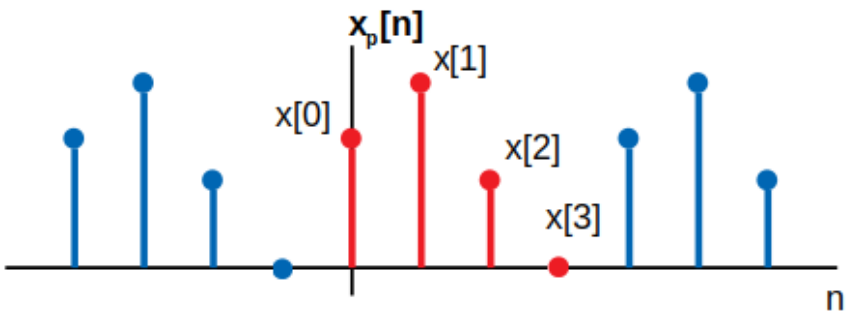
Now we analyze what happens when the periodic sequence $x_p[n]$ is shifted. For this we take the help of an example. Consider the signal $x[n]$ and its periodic extension $x_p[n]$ of period $N = 4$.

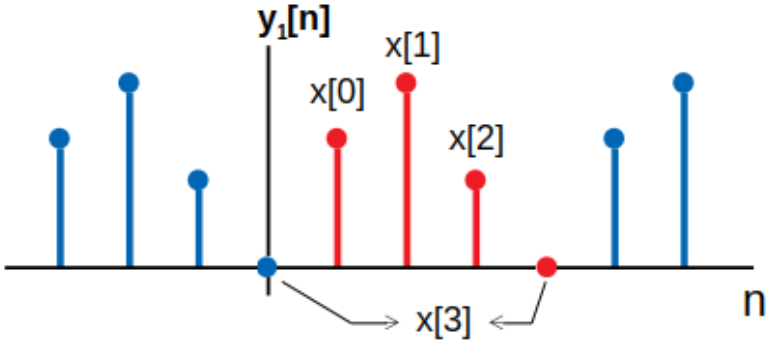




Circular symmetries

Right-shift by 1





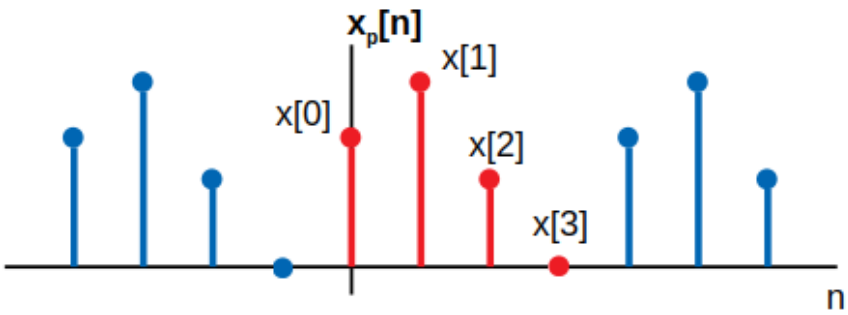
Note that

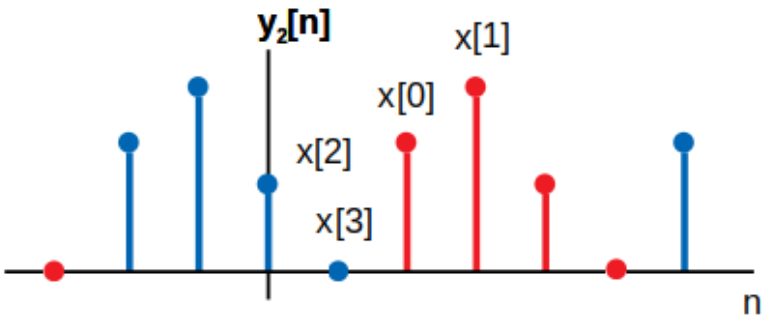
$$
\begin{aligned}
n = 0: &\quad y_1[0] = x_p[(-1)\,\mathtt{mod}\,4] &&\Rightarrow y_1[0] = x_p[3] \\
n = 1: &\quad y_1[1] = x_p[(0)\,\mathtt{mod}\,4] &&\Rightarrow y_1[1] = x_p[0] \\
n = 2: &\quad y_1[2] = x_p[(1)\,\mathtt{mod}\,4] &&\Rightarrow y_1[2] = x_p[1] \\
n = 3: &\quad y_1[3] = x_p[(2)\,\mathtt{mod}\,4] &&\Rightarrow y_1[3] = x_p[2]
\end{aligned}
$$

$$y_1[n] = x_p[(n-1)\,\mathtt{mod}\,4].$$

Circular symmetries

Right-shift by 2





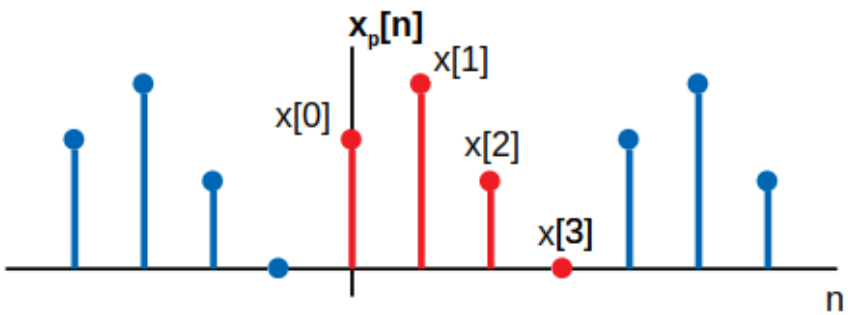
Note that

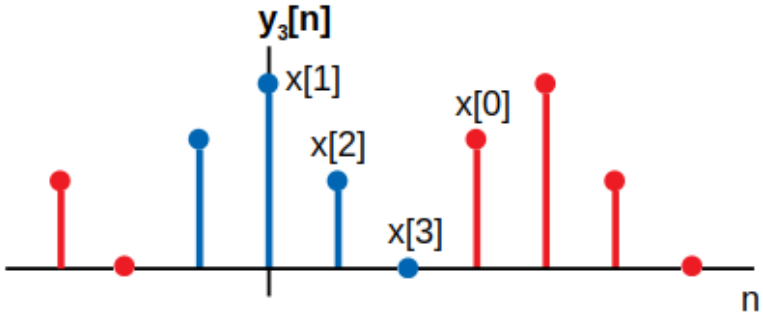
$$
\begin{aligned}
n = 0: &\quad y_2[0] = x_p[(-2)\,\mathtt{mod}\,4] &&\Rightarrow y_1[0] = x_p[2] \\
n = 1: &\quad y_2[1] = x_p[(-1)\,\mathtt{mod}\,4] &&\Rightarrow y_1[1] = x_p[3] \\
n = 2: &\quad y_2[2] = x_p[(0)\,\mathtt{mod}\,4] &&\Rightarrow y_1[2] = x_p[0] \\
n = 3: &\quad y_2[3] = x_p[(1)\,\mathtt{mod}\,4] &&\Rightarrow y_1[3] = x_p[1]
\end{aligned}
$$

$$tby_1[n] = x_p[(n-2)\,\mathtt{mod}\,4].$$

Circular symmetries

Right-shift by 3





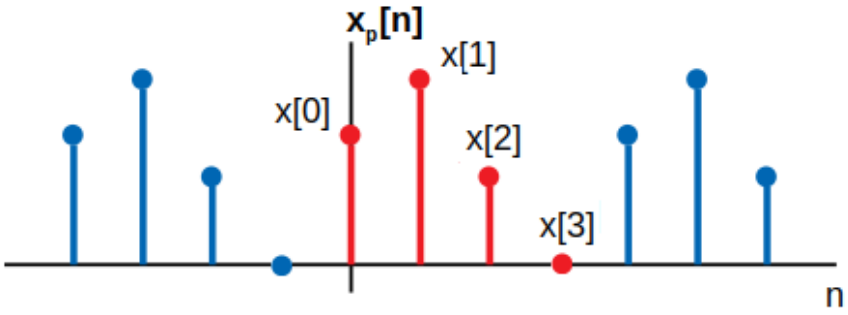
Note that

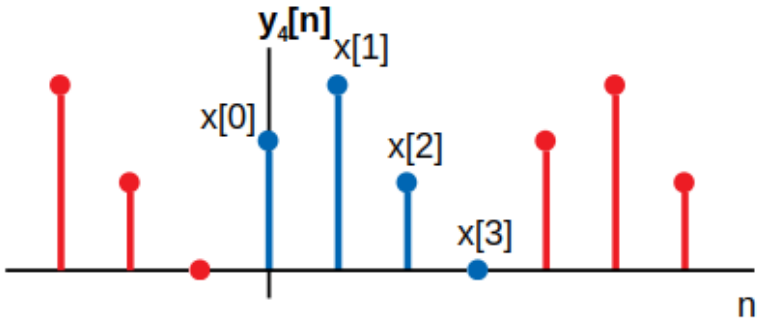
$$
\begin{aligned}
n = 0: &\quad y_2[0] = x_p[(-3) \bmod 4] &&\Rightarrow y_1[0] = x_p[1] \\
n = 1: &\quad y_2[1] = x_p[(-2) \bmod 4] &&\Rightarrow y_1[1] = x_p[2] \\
n = 2: &\quad y_2[2] = x_p[(-1) \bmod 4] &&\Rightarrow y_1[2] = x_p[3] \\
n = 3: &\quad y_2[3] = x_p[(0) \bmod 4] &&\Rightarrow y_1[3] = x_p[0]
\end{aligned}
$$

$$y_1[n] = x_p[(n-3) \bmod 4].$$

Circular symmetries

Right-shift by 4





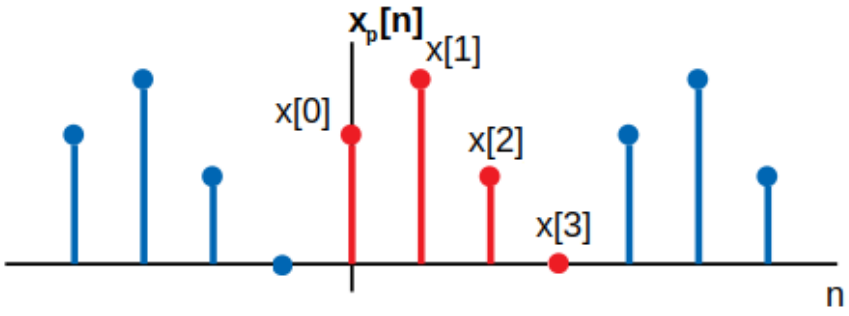
Note that

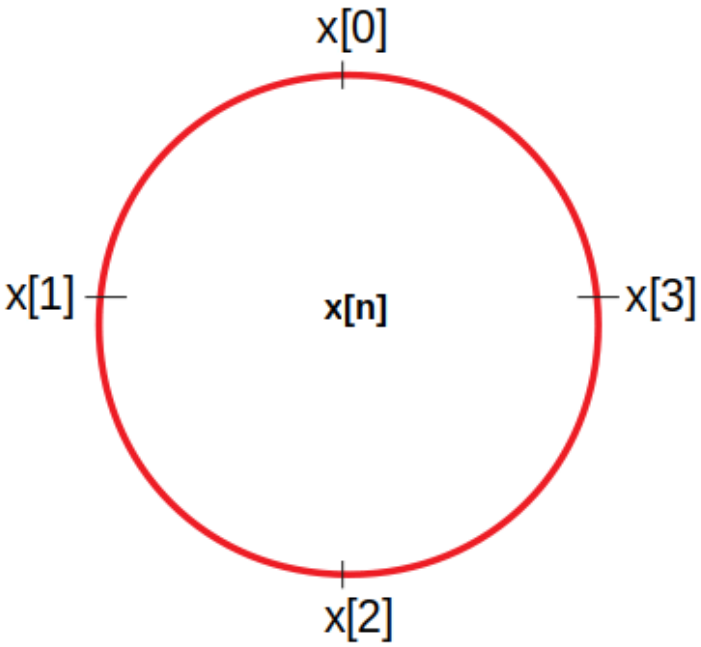
$$
\begin{aligned}
n = 0: &\quad y_2[0] = x_p[(-4)\,\mathtt{mod}\,4] &&\Rightarrow y_1[0] = x_p[0] \\
n = 1: &\quad y_2[1] = x_p[(-3)\,\mathtt{mod}\,4] &&\Rightarrow y_1[1] = x_p[1] \\
n = 2: &\quad y_2[2] = x_p[(-2)\,\mathtt{mod}\,4] &&\Rightarrow y_1[2] = x_p[2] \\
n = 3: &\quad y_2[3] = x_p[(-1)\,\mathtt{mod}\,4] &&\Rightarrow y_1[3] = x_p[3]
\end{aligned}
$$

$$y_1[n] = x_p[(n-4)\,\mathtt{mod}\,4] = x_p[n].$$

Circular symmetries

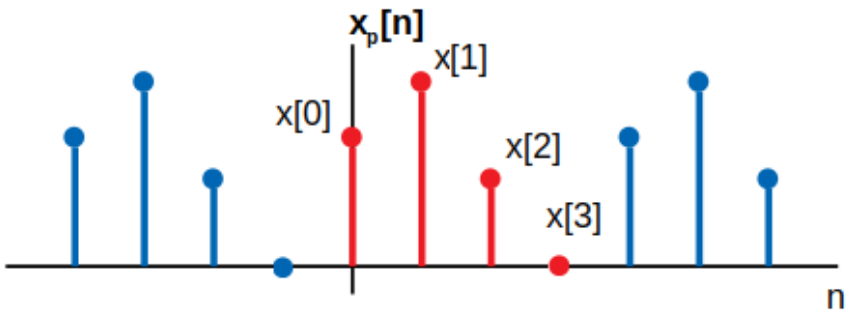
- In general, a periodic sequence $x_p[n]$ of period N right-shifted by k is given by $x_p[(n-k) \,\texttt{mod}\, N]$.
- Right shift operation of $x_p[n]$ can also be shown as a circular shift of $x[n]$. In order to do so we represent the signal $x[n]$ in a circle in this form:

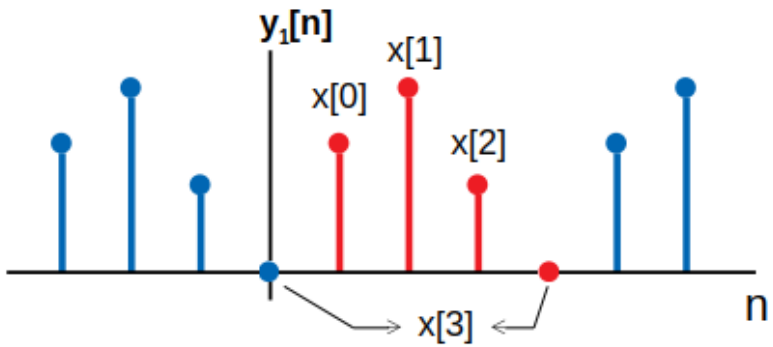


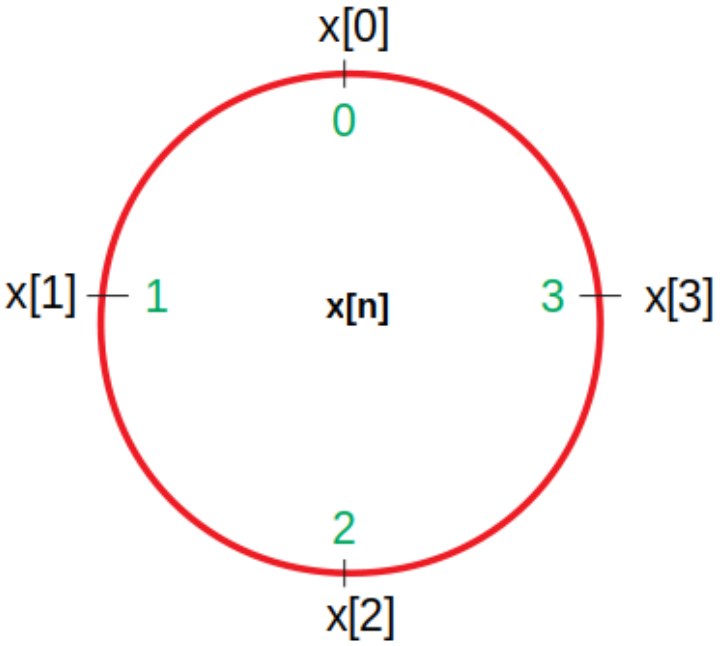


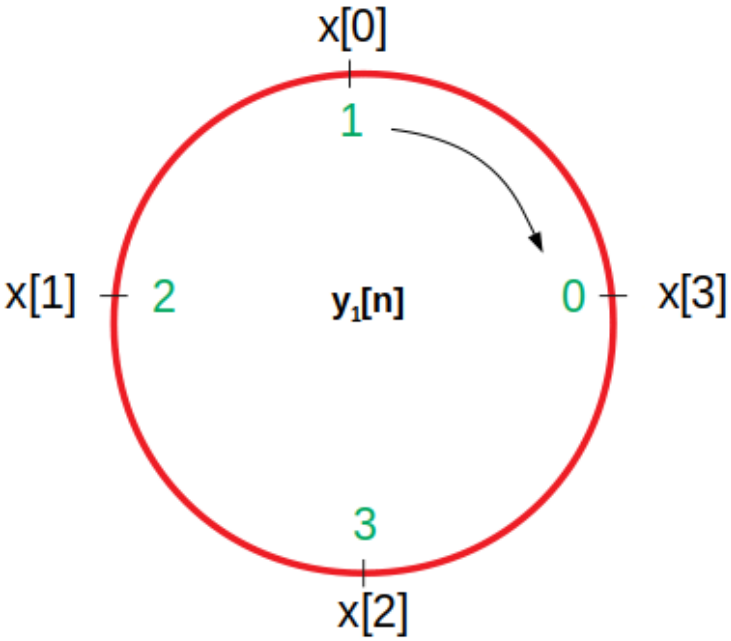
Circular symmetries

Right-shift by 1



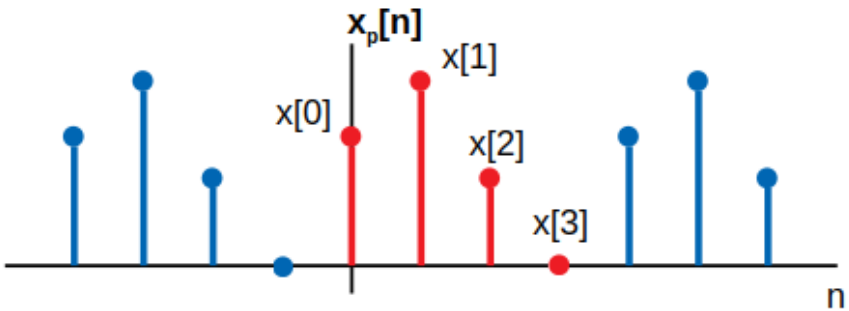


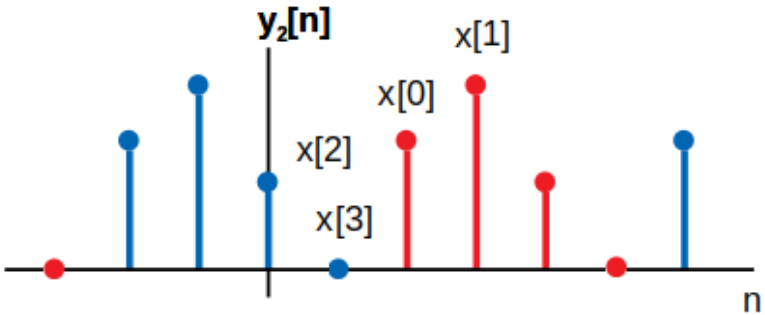


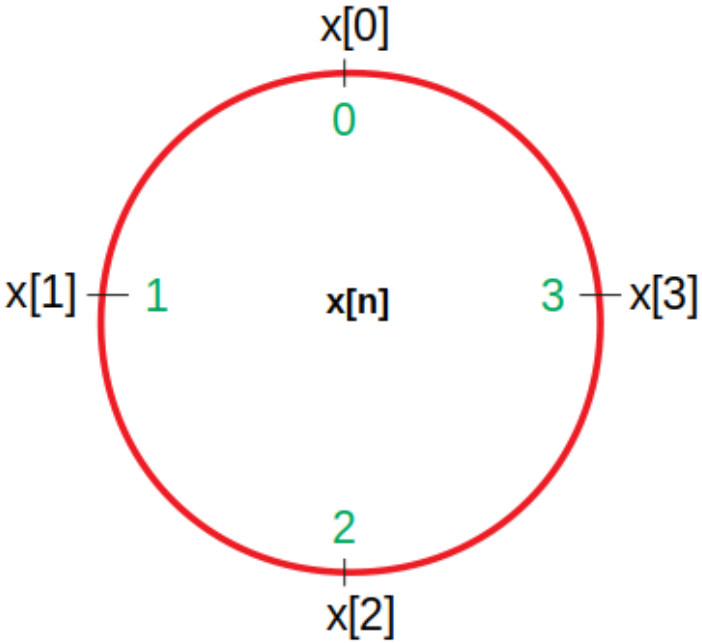


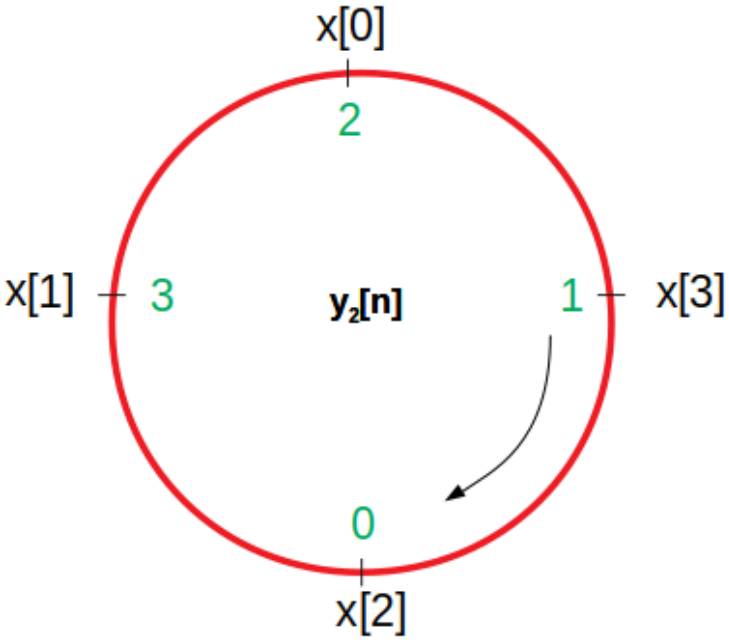
Circular symmetries

Right-shift by 2



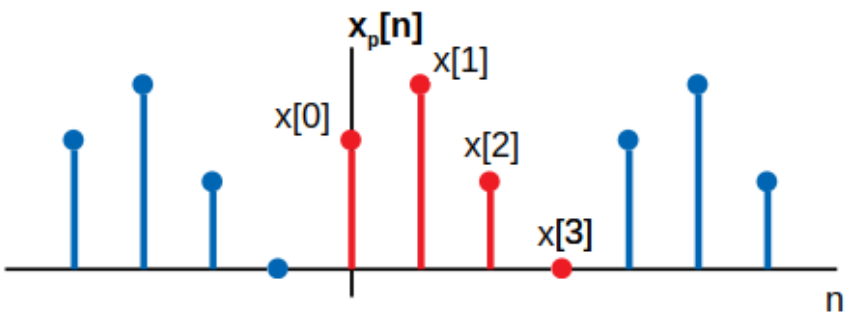


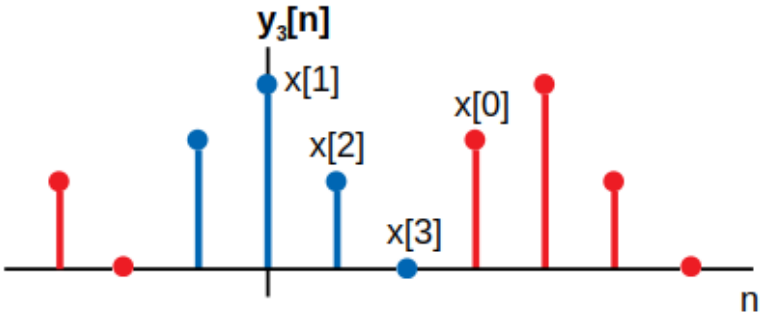


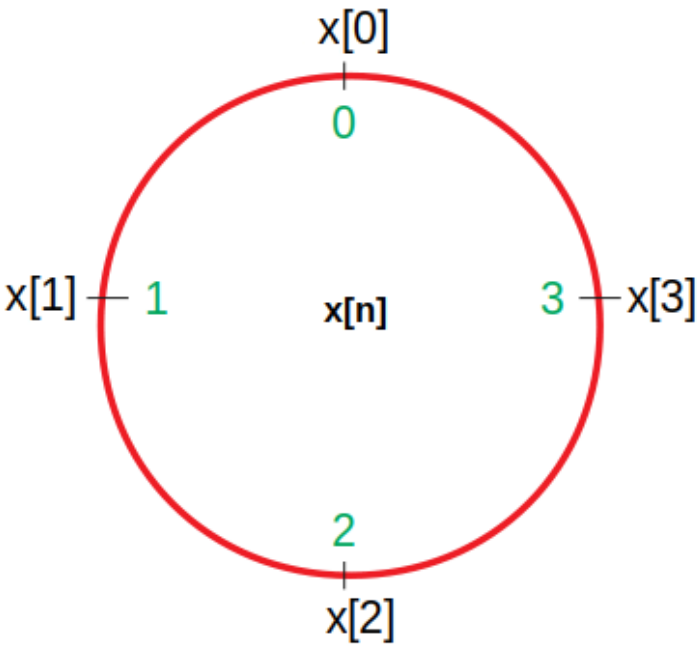


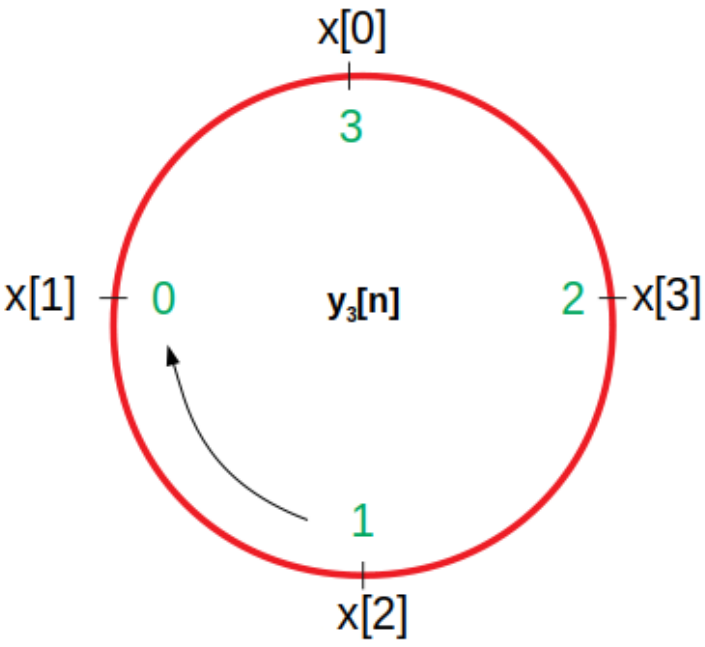
Circular symmetries

Right-shift by 3



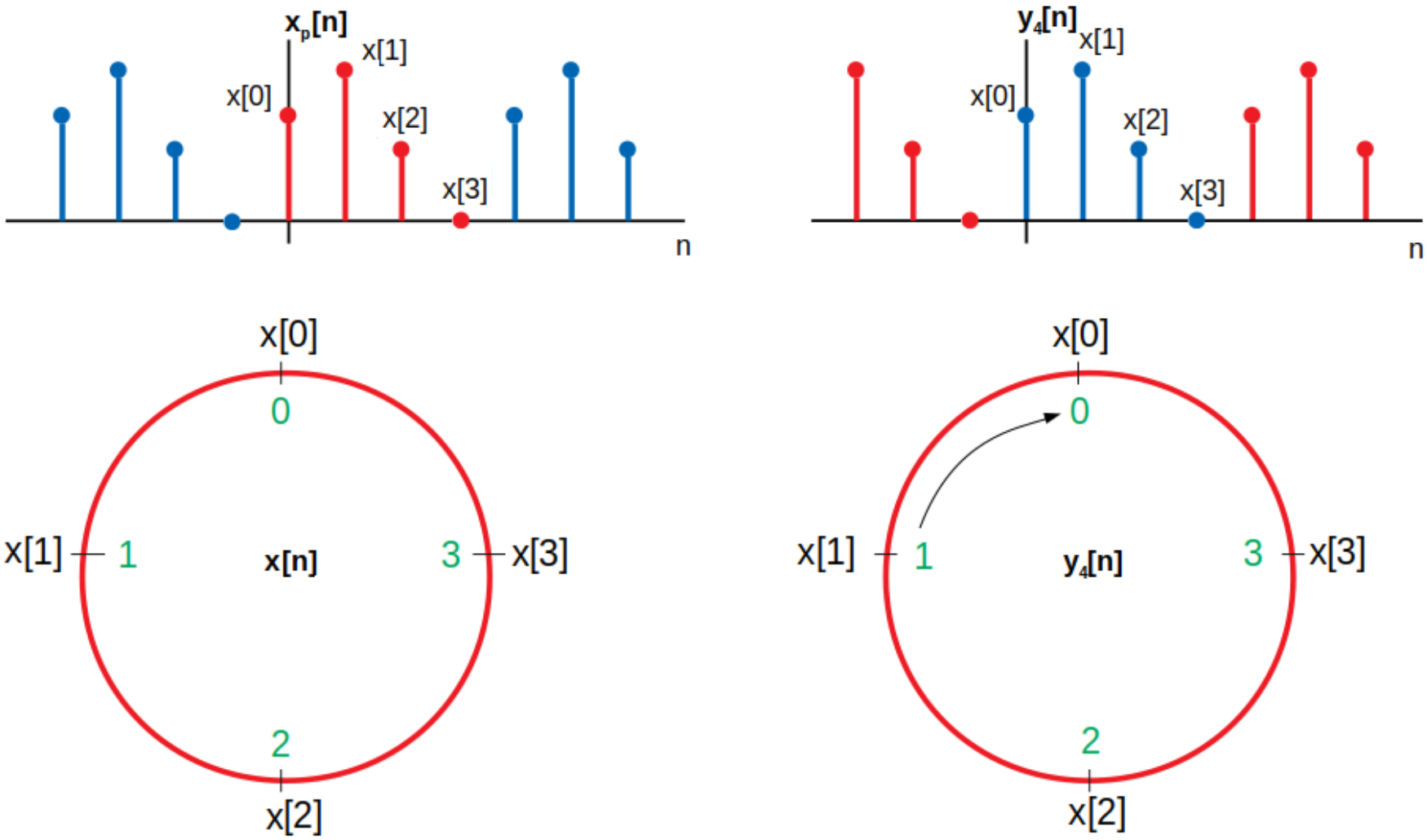






Circular symmetries

Right-shift by 4



Circular-shift of an N-point sequence is equivalent to a linear-shift of its periodic extension.

Circular symmetries

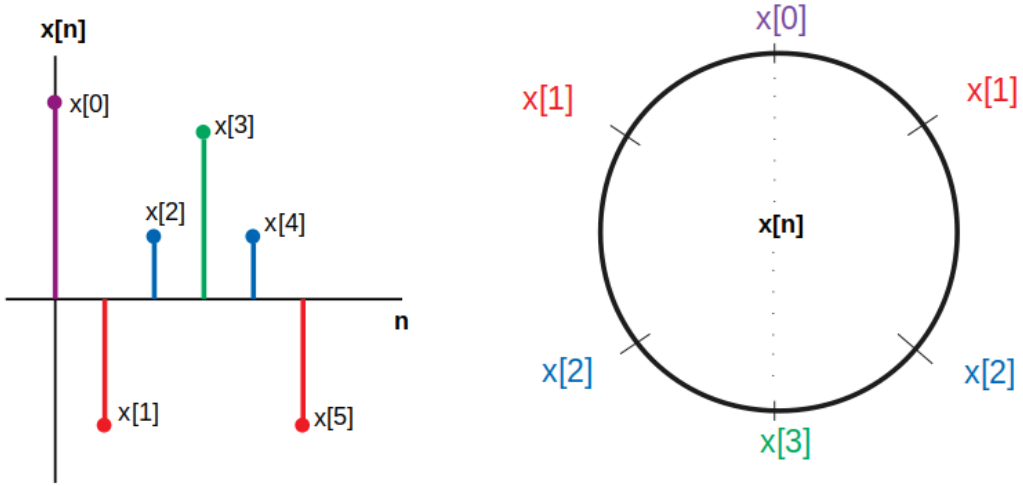
- The inherent periodicity resulting from the arrangement of the N-point sequence on the circumference of a circle dictates a different definition of even and odd symmetry, and time reversal of a sequence.

Definition

An N-point sequence is called *circularly even* if it is symmetric about the point zero on the circle. This implies that

$$x[N-n] = x[n], \qquad 1 \leqslant n \leqslant N-1.$$

Note: This is not 0



Circular symmetries

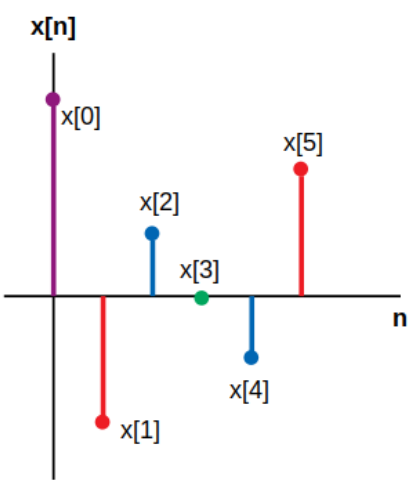
- The inherent periodicity resulting from the arrangement of the N-point sequence on the circumference of a circle dictates a different definition of even and odd symmetry, and time reversal of a sequence.

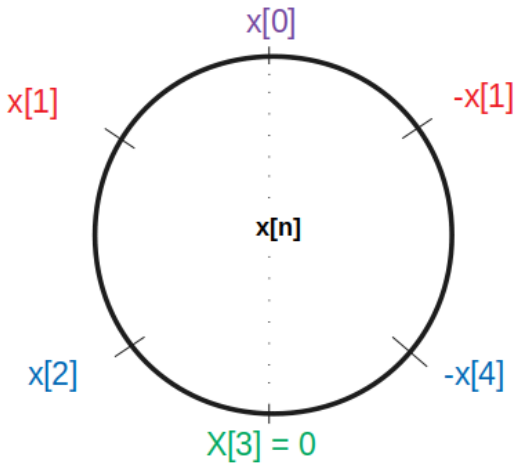
Definition

An N-point sequence is called *circularly odd* if it is antisymmetric about the point zero on the circle. This implies that

$$x[N-n] = -x[n], \qquad 1 \leqslant n \leqslant N-1.$$

Note: This is not 0

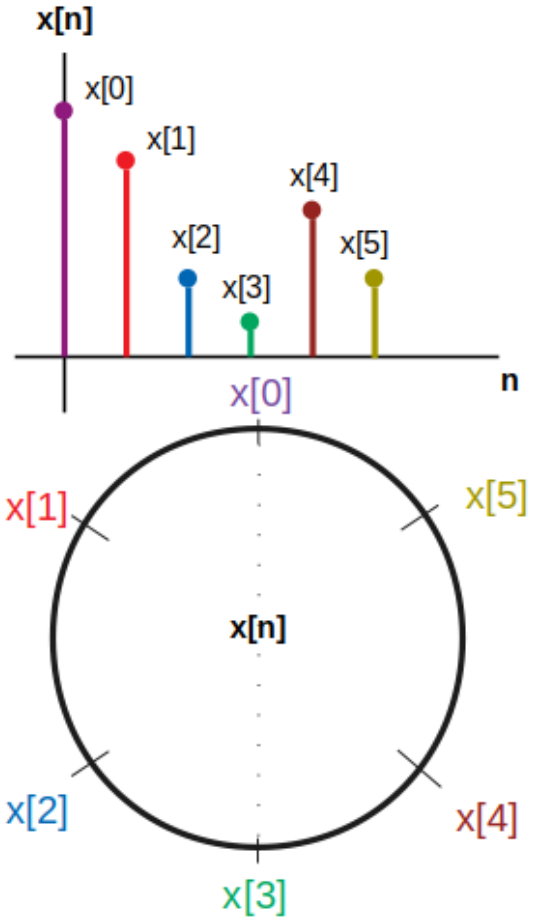


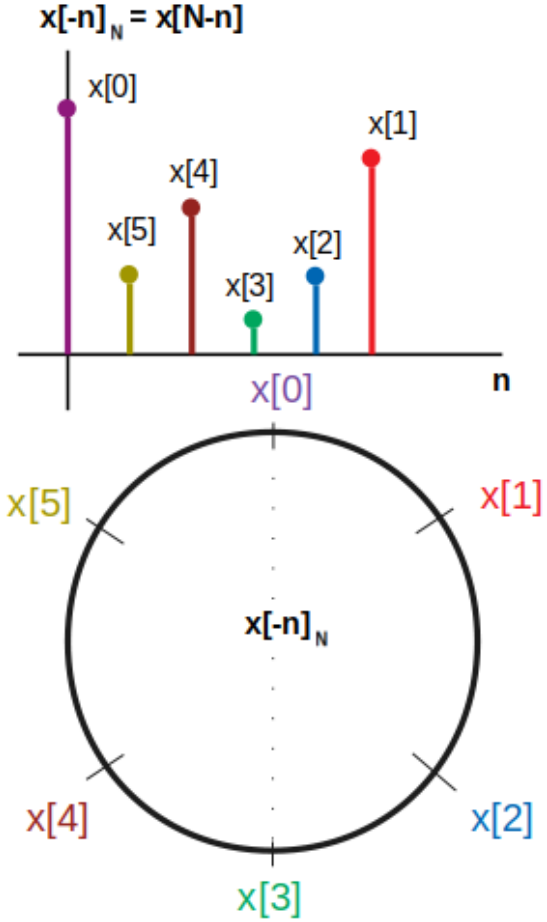


Time reversal

- Time reversal of an N-point sequence is obtained by reversing its samples about the point zero on the circle. Thus the sequence $x[-n]_N$ is given by

$$x[-n]_N = x[N-n], \qquad 0 \leqslant n \leqslant N-1.$$





Odd and even sequences

Now that we know about the circular symmetries and inverses, we can revisit the definitions of odd and even sequences associated with periodic sequences.

Even sequence

$x_p[n] = x_p[-n] = x_p[N-n].$

Odd sequence

$x_p[n] = -x_p[-n] = -x_p[N-n].$

If the periodic sequence is complex-valued, then

Conjugate even

$x_p[n] = x_p^*[N-n].$

Conjugate odd

$x_p[n] = -x_p^*[N-n].$

Using the relationships above we can decompose a periodic sequence into two parts:

$$x_p[n] = x_{pe}[n] + x_{po}[n], \tag{1}$$

where

$$x_{pe}[n] = \frac{1}{2}\left(x_p[n] + x_p^*[N-n]]\right) \quad \text{and} \quad x_{po}[n] = \frac{1}{2}\left(x_p[n] - x_p^*[N-n]\right)$$

Symmetry properties of DFT

Assume that $x[n]$ and the corresponding $X[k]$ are both complex valued. Then,

$$x[n] = x_{\mathtt{R}}[n][n] + jx_{\mathtt{I}}[n][n] \qquad 0 \leqslant n \leqslant N-1. \tag{2}$$

$$X[k] = X_{\mathtt{R}}[k][n] + jX_{\mathtt{I}}[k][n] \qquad 0 \leqslant k \leqslant N-1. \tag{3}$$

Using equation (2) in the expression for DFT:

$$\begin{aligned}
X_{\mathtt{R}}[k] + jX_{\mathtt{I}}[k] &= \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} \\
&= \sum_{n=0}^{N-1} \left(x_{\mathtt{R}}[n] + jx_{\mathtt{I}}[n]\right)\left(\cos\left(\frac{2\pi}{N}kn\right) - j\sin\left(\frac{2\pi}{N}kn\right)\right) \\
\Rightarrow X_{\mathtt{R}}[k] + jX_{\mathtt{I}}[k] &= \sum_{n=0}^{N-1}\left(x_{\mathtt{R}}[n]\cos\left(\frac{2\pi}{N}kn\right) + x_{\mathtt{I}}[n]\sin\left(\frac{2\pi}{N}kn\right)\right) \\
&\quad + j\sum_{n=0}^{N-1}\left(-x_{\mathtt{R}}[n]\sin\left(\frac{2\pi}{N}kn\right) + x_{\mathtt{I}}[n]\cos\left(\frac{2\pi}{N}kn\right)\right).
\end{aligned} \tag{4}$$

Symmetry properties of DFT

Assume that $x[n]$ and the corresponding $X[k]$ are both complex valued. Then,

$$x[n] = x_{\mathtt{R}}[n][n] + jx_{\mathtt{I}}[n][n] \qquad 0 \leqslant n \leqslant N-1. \tag{2}$$

$$X[k] = X_{\mathtt{R}}[k][n] + jX_{\mathtt{I}}[k][n] \qquad 0 \leqslant k \leqslant N-1. \tag{3}$$

Similarly, Using equation (3) in the expression for IDFT:

$$\begin{aligned}
x_{\mathtt{R}}[n] + jx_{\mathtt{I}}[n] &= \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn} \\
&= \frac{1}{N}\sum_{k=0}^{N-1} \left(X_{\mathtt{R}}[k] + jX_{\mathtt{I}}[k]\right)\left(\cos\left(\frac{2\pi}{N}kn\right) + j\sin\left(\frac{2\pi}{N}kn\right)\right) \\
\Rightarrow x_{\mathtt{R}}[n] + jx_{\mathtt{I}}[n] &= \frac{1}{N}\sum_{k=0}^{N-1}\left(X_{\mathtt{R}}[k]\cos\left(\frac{2\pi}{N}kn\right) - X_{\mathtt{I}}[k]\sin\left(\frac{2\pi}{N}kn\right)\right) \\
&\quad + j\frac{1}{N}\sum_{k=0}^{N-1}\left(X_{\mathtt{R}}[k]\sin\left(\frac{2\pi}{N}kn\right) + X_{\mathtt{I}}[k]\cos\left(\frac{2\pi}{N}kn\right)\right).
\end{aligned} \tag{5}$$

Symmetry properties of DFT

Assume that $x[n]$ and the corresponding $X[k]$ are both complex valued. Then,

$$x[n] = x_{\mathtt{R}}[n][n] + jx_{\mathtt{I}}[n][n] \qquad 0 \leqslant n \leqslant N-1. \tag{2}$$

$$X[k] = X_{\mathtt{R}}[k][n] + jX_{\mathtt{I}}[k][n] \qquad 0 \leqslant k \leqslant N-1. \tag{3}$$

From equation (4) and equation (5):

$$X_{\mathtt{R}}[k] = \sum_{n=0}^{N-1} \left(x_{\mathtt{R}}[n] \cos\left(\frac{2\pi}{N}kn\right) + x_{\mathtt{I}}[n] \sin\left(\frac{2\pi}{N}kn\right)\right) \tag{6}$$

$$X_{\mathtt{I}}[k] = \sum_{n=0}^{N-1} \left(-x_{\mathtt{R}}[n] \sin\left(\frac{2\pi}{N}kn\right) + x_{\mathtt{I}}[n] \cos\left(\frac{2\pi}{N}kn\right)\right) \tag{7}$$

$$x_{\mathtt{R}}[n] = \frac{1}{N}\sum_{k=0}^{N-1} \left(X_{\mathtt{R}}[k] \cos\left(\frac{2\pi}{N}kn\right) - X_{\mathtt{I}}[k] \sin\left(\frac{2\pi}{N}kn\right)\right) \tag{8}$$

$$x_{\mathtt{I}}[n] = \frac{1}{N}\sum_{k=0}^{N-1} \left(X_{\mathtt{R}}[k] \sin\left(\frac{2\pi}{N}kn\right) + X_{\mathtt{I}}[k] \cos\left(\frac{2\pi}{N}kn\right)\right) \tag{9}$$

Multiplication of DFT

$$x_1[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_1[k] \qquad \text{and} \qquad x_2[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_2[k]$$

$$\boxed{x_1[n] \otimes x_2[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_1[k]X_2[k]}$$

Proof: Let $X_3[k] := X_1[k]X_2[k]$. Let the IDFT of $X_3[k]$ be $x_3[n]$. Then,

$$\begin{aligned}
x_3[n] &= \frac{1}{N}\sum_{k=0}^{N-1} X_3[k]e^{j\frac{2\pi}{N}nk} = \frac{1}{N}\sum_{k=0}^{N-1} X_1[k]X_2[k]e^{j\frac{2\pi}{N}nk} \\
&= \frac{1}{N}\sum_{k=0}^{N-1}\left(\sum_{m=0}^{N-1} x_1[m]e^{-j\frac{2\pi}{N}km}\right)\left(\sum_{\ell=0}^{N-1} x_2[\ell]e^{-j\frac{2\pi}{N}k\ell}\right) \\
&= \frac{1}{N}\sum_{m=0}^{N-1} x_1[m]\sum_{\ell=0}^{N-1} x_2[\ell]\left(\sum_{k=0}^{N-1} e^{j\frac{2\pi}{N}(m-n-\ell)k}\right)
\end{aligned}$$

We analyze the part in red now.

Multiplication of DFT

$$x_1[n] \xleftrightarrow[N]{DFT} X_1[k] \qquad \text{and} \qquad x_2[n] \xleftrightarrow[N]{DFT} X_2[k]$$

$$\boxed{x_1[n] \otimes x_2[n] \xleftrightarrow[N]{DFT} X_1[k]X_2[k]}$$

Proof: Let $a := e^{j\frac{2\pi}{N}(m-n-\ell)}$. Then note that

$$\sum_{k=0}^{N-1} e^{j\frac{2\pi}{N}(m-n-\ell)k} = \sum_{k=0}^{N-1} a^k = \begin{cases} N, & a = 1 \\ \dfrac{1-a^N}{1-a}, & a \neq 1. \end{cases} \tag{10}$$

Observe that $a = 1$ when $n - m - \ell$ is a multiple of N. On the other hand, $a^N = 1$ for any value of $a \neq 0$. Therefore, equation (10) reduces to

$$\sum_{k=0}^{N-1} a^k = \sum_{k=0}^{N-1} e^{j\frac{2\pi}{N}(m-n-\ell)} = \begin{cases} N, & \ell = (n-m) \,\texttt{mod}\, N \text{ and } p \text{ is an integer} \\ 0, & \text{otherwise.} \end{cases}$$

Multiplication of DFT

$$x_1[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_1[k] \qquad \text{and} \qquad x_2[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_2[k]$$

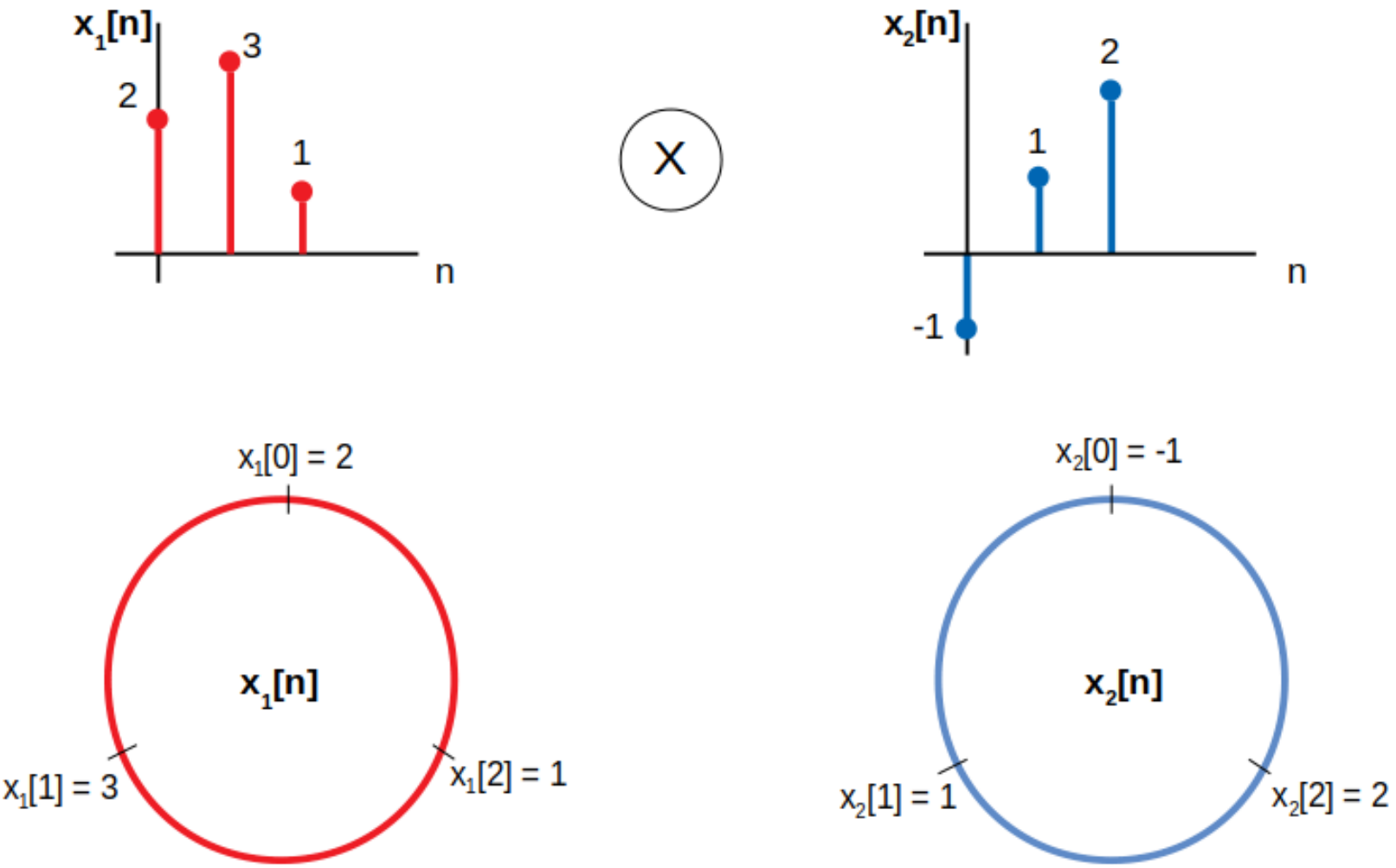
$$x_1[n] \otimes x_2[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_1[k]X_2[k]$$

Proof: Let $X_3[k] := X_1[k]X_2[k]$. Let the IDFT of $X_3[k]$ be $x_3[n]$. Then,

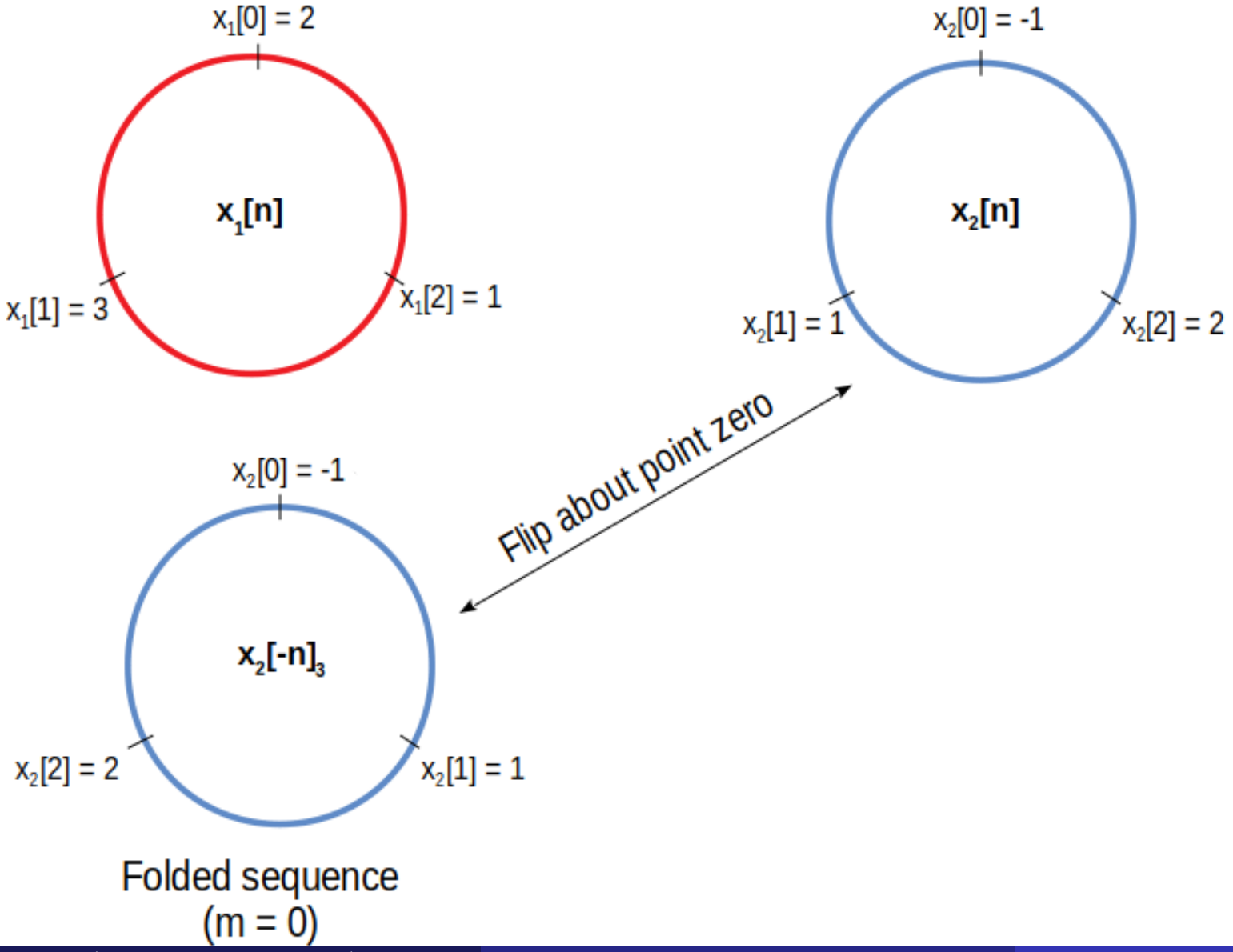
$$\begin{aligned} x_3[n] &= \frac{1}{N}\sum_{m=0}^{N-1} x_1[m] \sum_{\ell=0}^{N-1} x_2[\ell] \left(\sum_{k=0}^{N-1} e^{j\frac{2\pi}{N}(m-n-\ell)k} \right) \\ &= \sum_{m=0}^{N-1} x_1[m] x_2[n-m]_N, \quad m = 0, 1, \cdots, N-1. \end{aligned}$$

Multiplication in DFT results in circular convolution in the time domain. This is similar to the DTFS case.
We take an example next to illustrate the procedure to compute circular convolution pictorially.

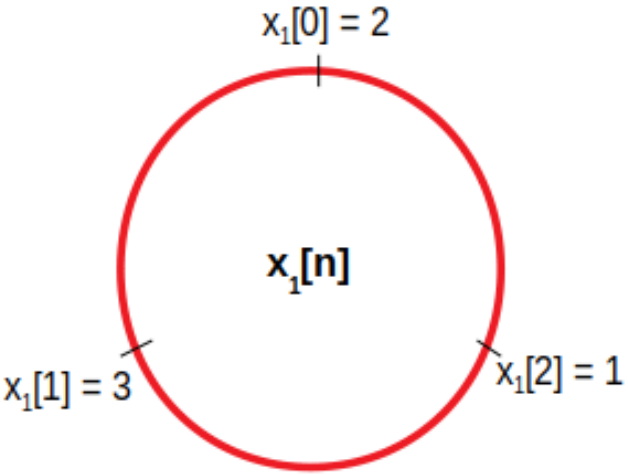
Circular convolution

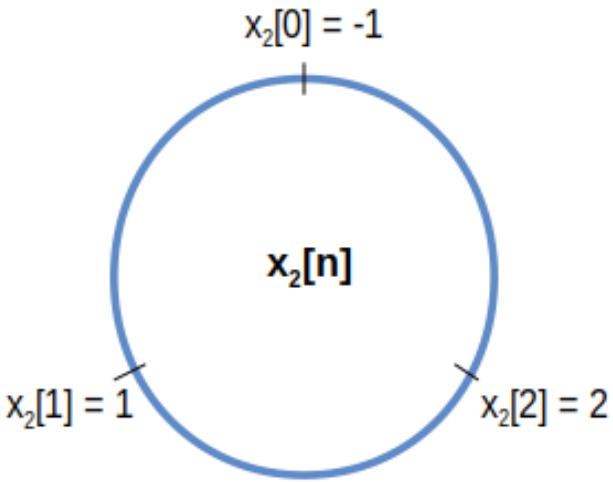


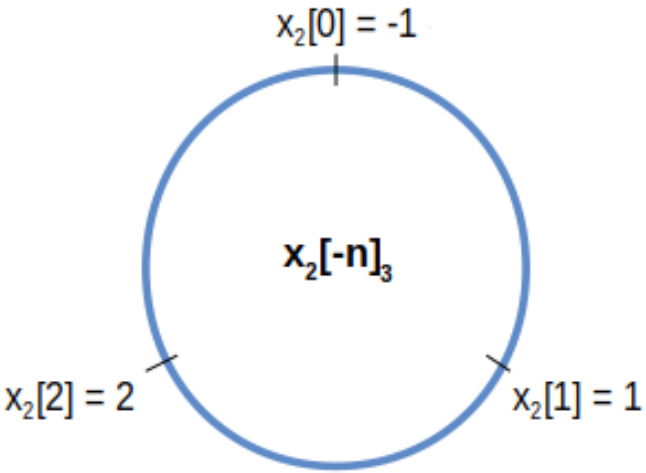
Circular convolution



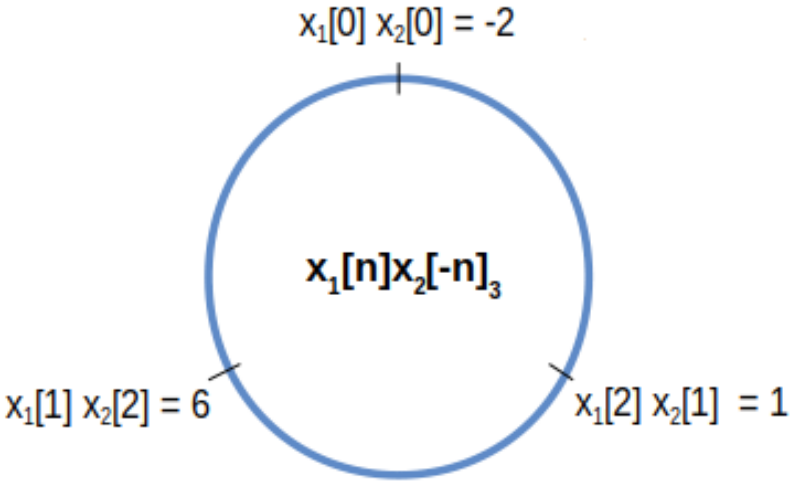
Circular convolution





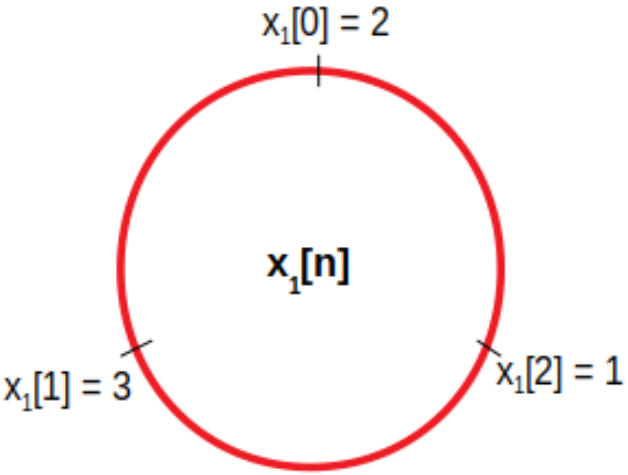


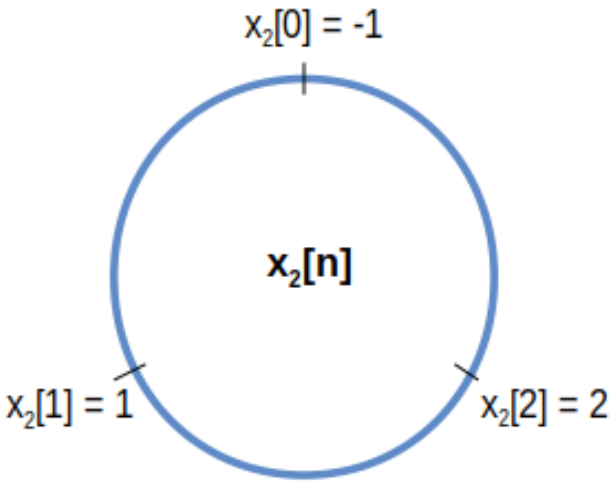
Folded sequence
(m = 0)

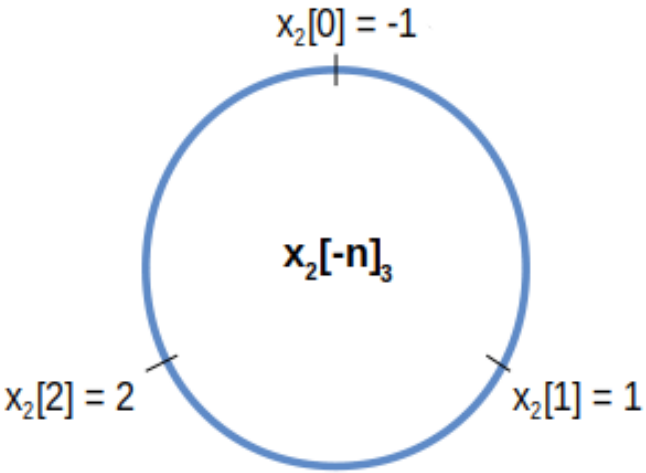


Product sequence

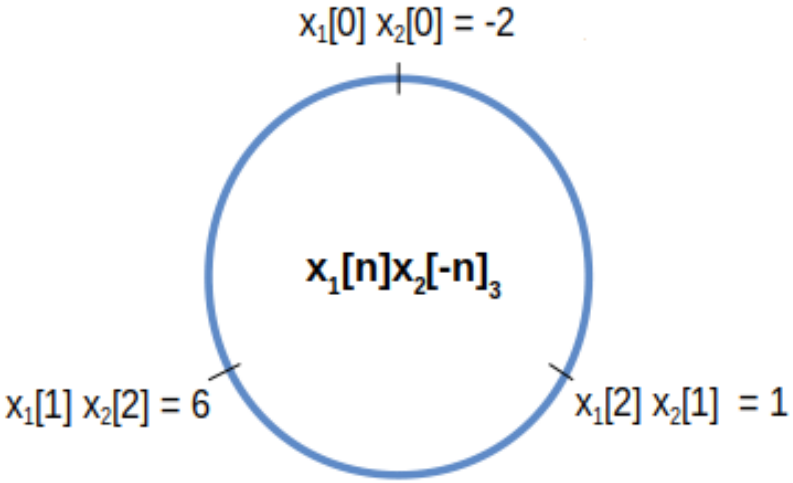
Circular convolution







Folded sequence
(m = 0)



Product sequence

$x_3[0] =$

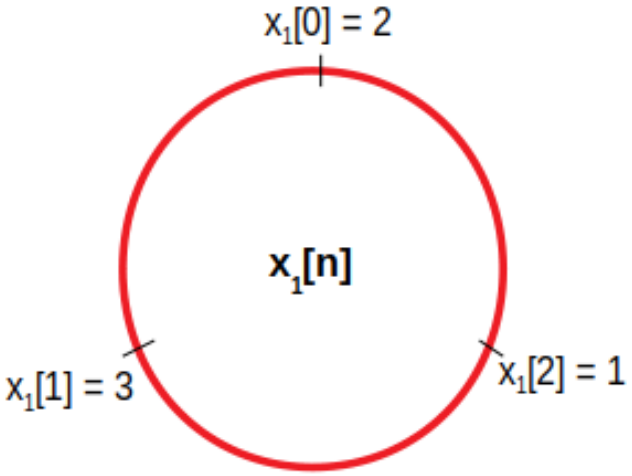
$x_1[0]\ x_2[0] +$

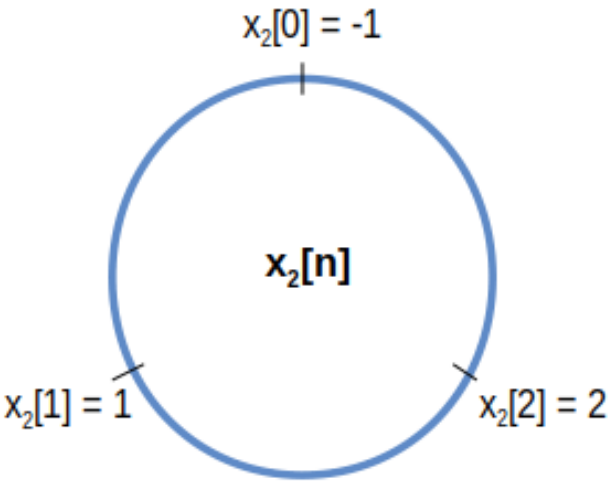
$x_1[1]\ x_2[2] +$

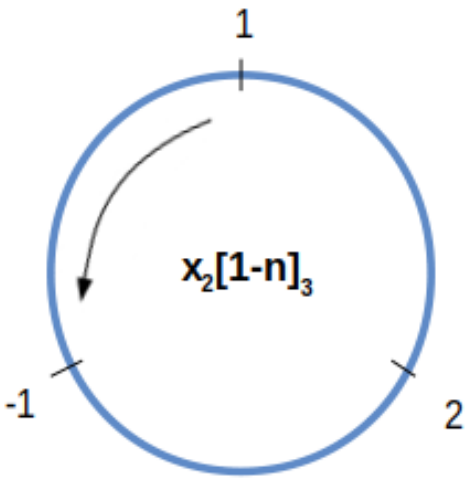
$x_1[2]\ x_2[1]$

$= 5$

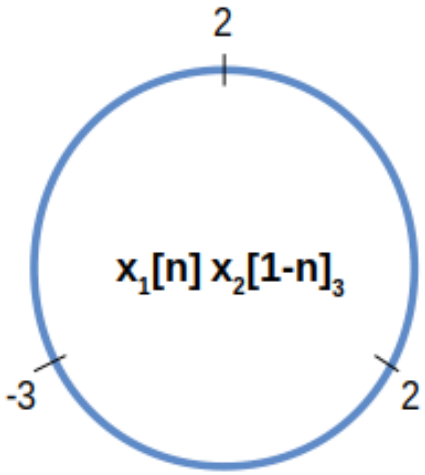
Circular convolution







Folded sequence rotated by one unit anticlockwise (m = 1)



Product sequence

$x_3[1] =$

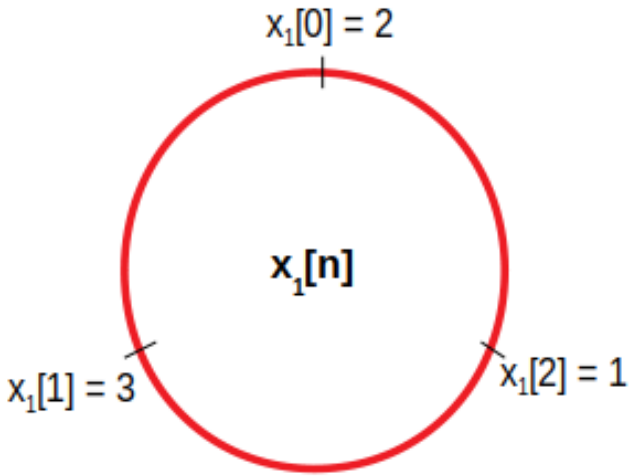
$x_1[0]\, x_2[1] +$

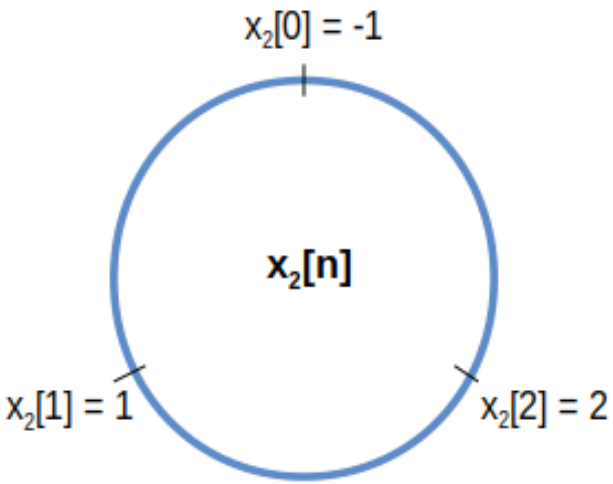
$x_1[1]\, x_2[0] +$

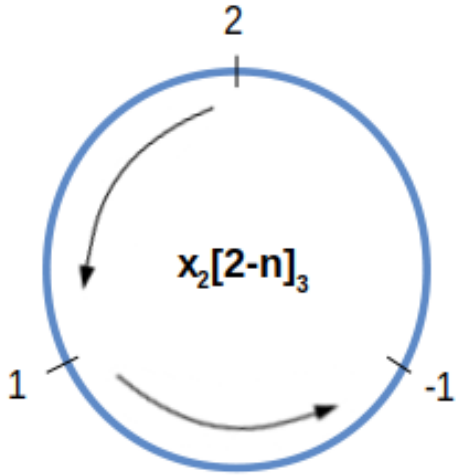
$x_1[2]\, x_2[2]$

$= 1$

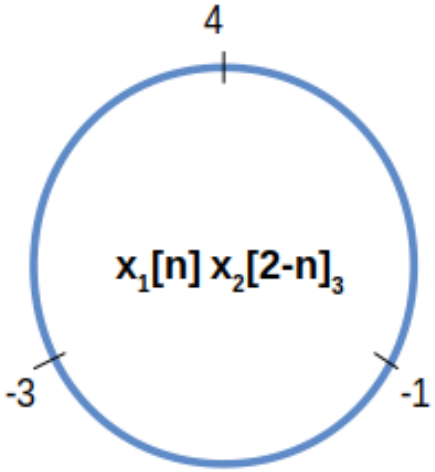
Circular convolution







Folded sequence rotated by two units anticlockwise (m = 2)



Product sequence

$x_3[2] =$

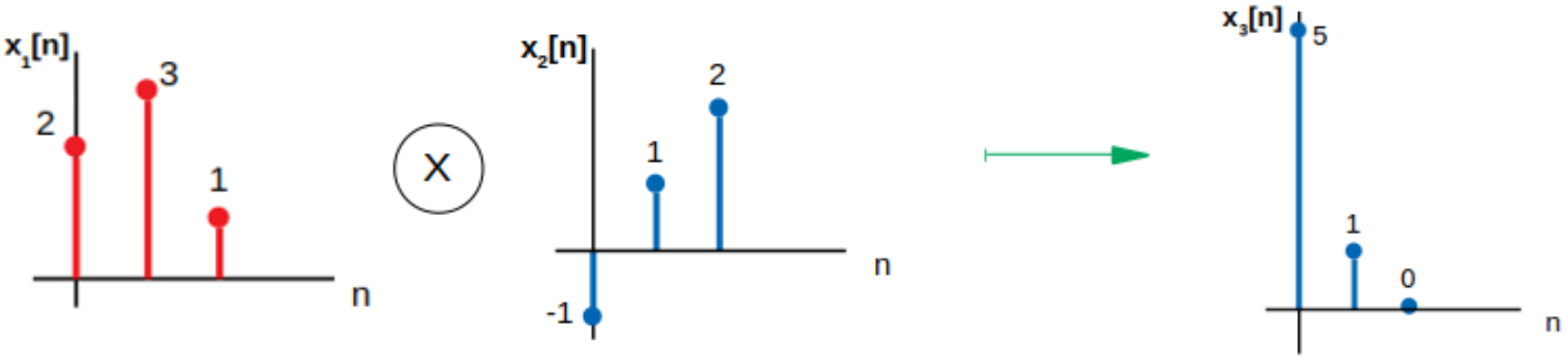
$x_1[0]\, x_2[2] +$

$x_1[1]\, x_2[1] +$

$x_1[2]\, x_2[0]$

$= 0$

Circular convolution



- Observe that circular convolution involves the same four steps as the ordinary linear convolution: *folding* (time reversing) one sequence, *shifting* the folded sequence, *multiplying* the two sequences to obtain the product sequence, *summing* the values of the product sequence.
- Basic difference between these two convolution is that, in circular convolution, the folding and shifting operations are performed in a circular fashion by computing the index of one of the sequences modulo N. In linear convolution there is no modulo N operation.

Other properties

$$x_1[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_1[k] \qquad \text{and} \qquad x_2[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_2[k]$$

Time reversal

$$x_1[-n]_N \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_1[k]_N = X_1[N-k]$$

Circular time shift

$$x_1[n-\ell]_N \overset{DFT}{\underset{N}{\Longleftrightarrow}} X_1[k]e^{-j\frac{2\pi}{N}k\ell}$$

Multiplication of sequences

$$x_1[n]x_2[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} \frac{1}{N}X_1[k] \otimes X_2[k].$$

Parseval's theorem

$$\sum_{n=0}^{N-1} x_1[n]x_2^*[n] \overset{DFT}{\underset{N}{\Longleftrightarrow}} \frac{1}{N}\sum_{k=0}^{N-1} X[k]X_2^*[k].$$